BITS Pilani
Pilani Campus

Operation Management

Abhishek Maiti

innovate achieve lead

BITS Pilani
Pilani Campus

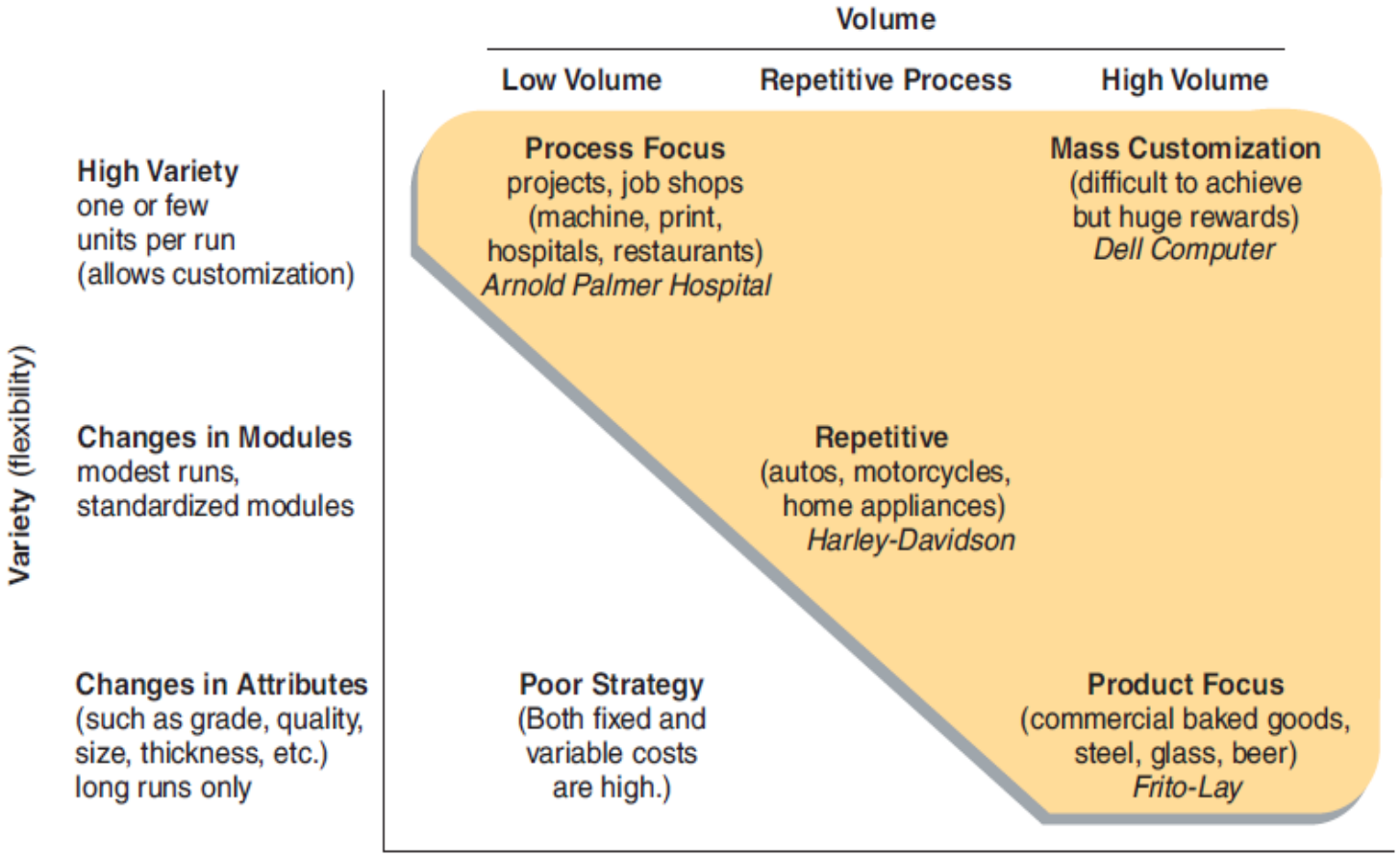
<MBACC526, Operation Management>
Lecture No. 4

Process Strategy

- An organization's approach to transform resources into goods and services.
- The strategy to manage the efficiency and flexibility of production as well as cost and quality of the goods produced.

Different types of Process Strategy

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management



Source: Heizer, Render, Muckson

Process Focus

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

- Small Quantity and Large Variety of products.
- Broadly skilled operators.
- Instruction for each job.
- High inventory.
- Finished goods are made to order.
- Complex Scheduling.
- Low FC and High VC

Repetitive Focus

- Long Runs and Standardized product from modules.
- Moderately trained employees.
- Few changes in job instruction.
- Finished goods are made to frequent forecast.
- Routine Scheduling.

Product Focus

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

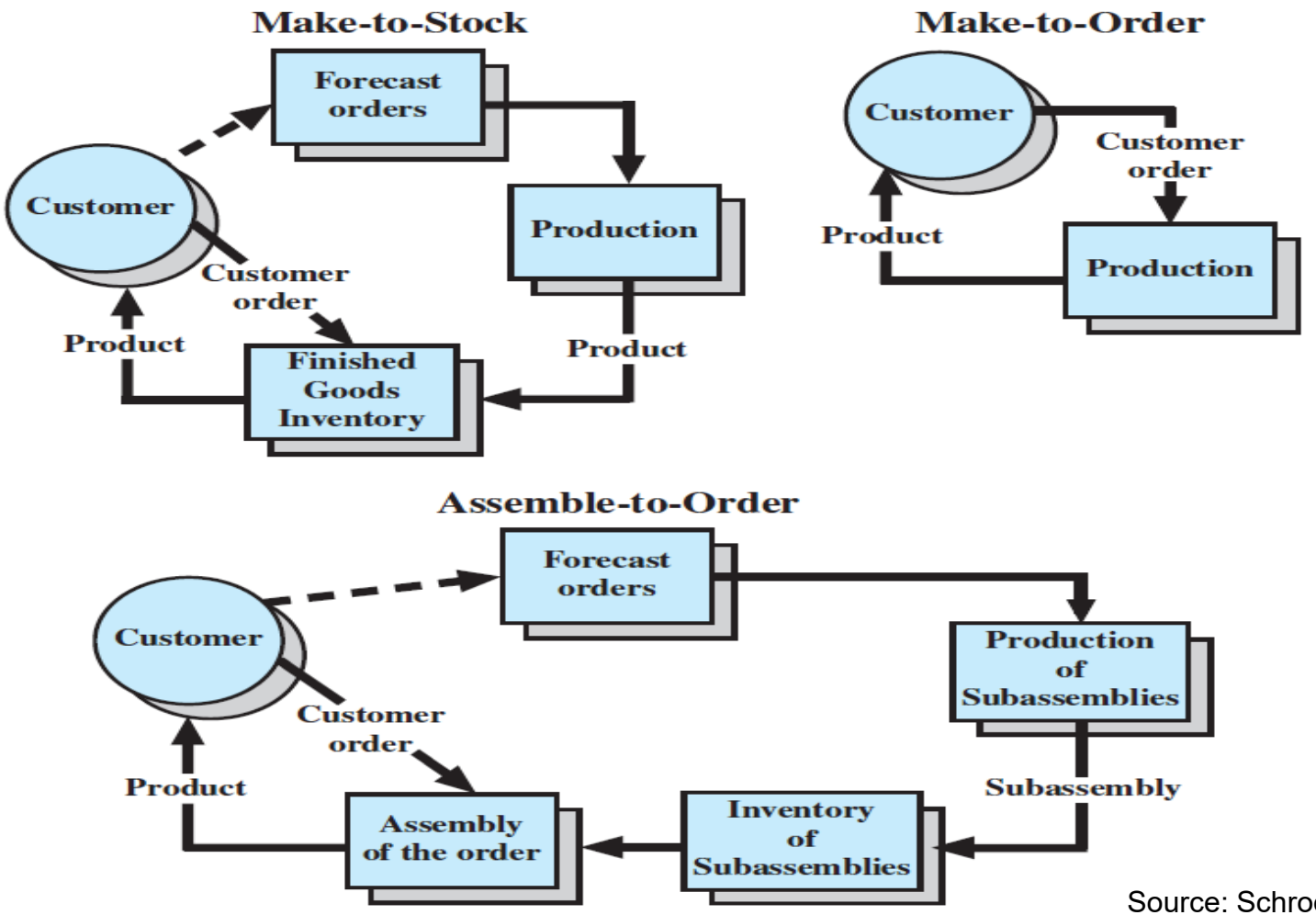
- Large Quantity and small variety of products.
- Less broadly skilled operators.
- Standardized job instructions.
- Low inventory.
- Finished goods are

made to forecast and store.

- Routine Scheduling.
- High FC and low VC

Mass Customization

- Large Quantity and large variety of products
- Flexible operators.
- Custom order requiring many job instructions.
- Low inventory relative to the value of the product
- Finished goods are build to Order (BTO)
- Sophisticated Scheduling accommodates custom order.
- Fixed cost tend to be high and variable costs low.



Source: Schroeder, Goldstein

Case let: Subway

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

Operations Leader SUBWAY Uses Assemble-to-Order

There are more than 44,000 SUBWAY restaurants in 111 countries around the world. The way they produce and serve food is an example of how a firm uses an assemble-to-order process.

Some sandwich ingredients, such as the bread, meatballs, and sauces, are made to stock as subassemblies. These items are produced either at individual SUBWAY restaurants or by suppliers and then held in stock at the restaurants. Batch processes are used to make these items as efficiently as possible while ensuring that standards of quality are met.

© Tim Boyle/Getty Images

When a customer comes in to order a sandwich, an ATO process is used. The customer specifies the type of bread to use as well as each of the sandwich ingredients. Each sandwich can be customized to meet customer requests exactly. This is the advantage of an ATO process.

Using an ATO process ensures that SUBWAY operates both efficiently and effectively. Some food is produced in batches, but the final product—the sandwich the customer receives—is produced one at a time.

Source: www.subway.com, 2016.

Any Question

Process Comparison

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

- Breakeven Analysis :

$$\text{Demand} = \frac{Fixed\ Cost}{Price - Variable\ Cost}$$

- Crossover Chart : A chart of costs at the possible volumes for more than one process.
- Focussed Process : To build core competence that fosters market and financial success.

Breakeven Analysis

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

Travis and Jeff own Up Right Paddlers, a new start-up company with the goal of designing, making, and marketing stand-up paddle boards for streams and rivers. It is alternative to Surfboards.

Unlike surfboards that market for $500 to $1000 each, paddle boards are typically sold for between $100 and $400. Since Travis and Jeff are just starting out and the demand for paddle boards on the East Coast has not been firmly established, they anticipate selling their product for $100 each. Travis estimates the fixed cost for equipment and space will be $2000 and the material and labor costs will run $50 per unit. What volume of demand will be necessary for Travis and Jeff to break even on their new venture?

- Jeff, the more optimistic of the two owners of UpRight Paddlers, believes that demand forpaddle boards will exceed the breakeven point of 40 units He proposes spending $10,000 in fixed costs to buy more automated equipment that would reduce the materials and labor cost to $30 per board. The boards would sell for $100, regardless of which manufacturing process is chosen.
- Compare the two processes and determine for what level of demand each process would be preferred. Label Travis' proposal as Process A, and Jeff's proposal as Process B.

Crossover Chart

- Kleber Enterprises would like to evaluate three accounting software products (A, B, and C) to support changes in its internal accounting processes. The resulting processes will have cost structures similar to
- The costs of the software for these processes are:

	Total Fixed Cost	Dollar Required per Accounting Report
Software A	$200,000	$60
Software B	$300,000	$25
Software C	$400,000	$10

Any Question

Focused Process

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

Customers .

Products.

Focused Process

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

Service

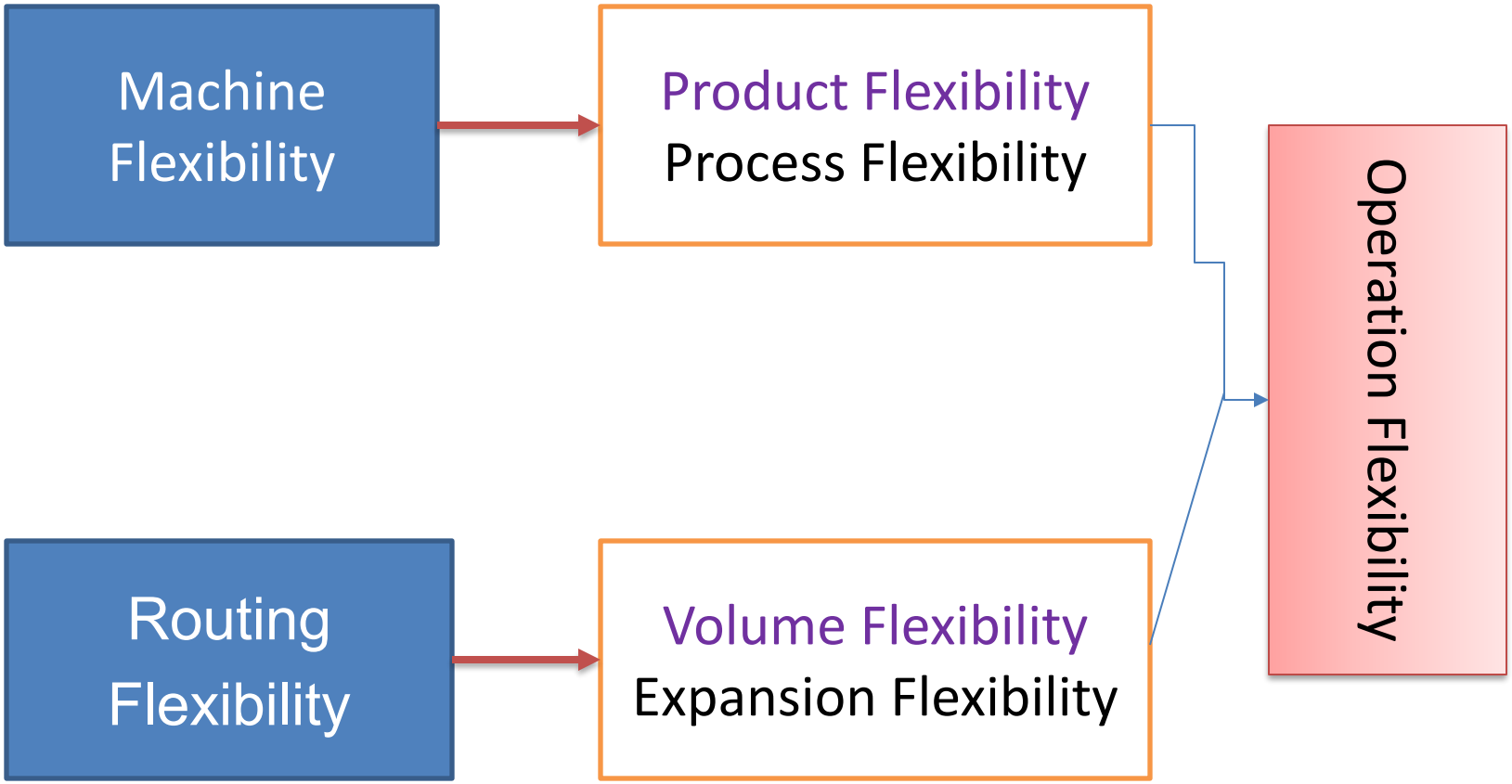
Technology

Flexibility

- The ability to respond with little penalty in time, cost, or customer value.

Operation Flexibility (FMS)

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management



Components of FMS

- Workstations.
- Material handling and storage system.
- Computer control system.
- People.

Suitability of FMS

FMS is best suited when

–all products use the same family of components;

–the number of components is moderate.

–the volume of each component is also moderate.

–Small lot size.

Benefits of FMS

- Inventory reduction of 60-80 per cent.
- Direct labour savings of 30-50 per cent.
- Increased asset usage approaching 80-90 per cent.
- Floor space reduction of 40-50 per cent.

Any Question

Process Analysis & Design

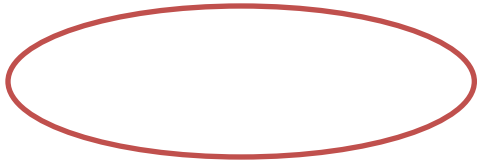
- The goal is to achieve competitive advantage in terms of low cost, differentiation and response.
- It eliminates the NVA, maximizes CLV and become order winners.

Process Analysis & Design Tools

- Flowchart.
- Time Function Mapping.
- Process Chart.
- Value Stream Mapping.

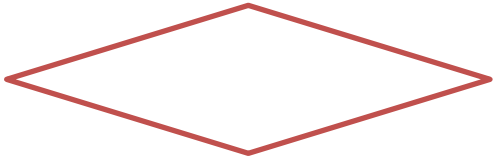
Flow Chart

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management



Start Point & End Point

Sequence of Steps

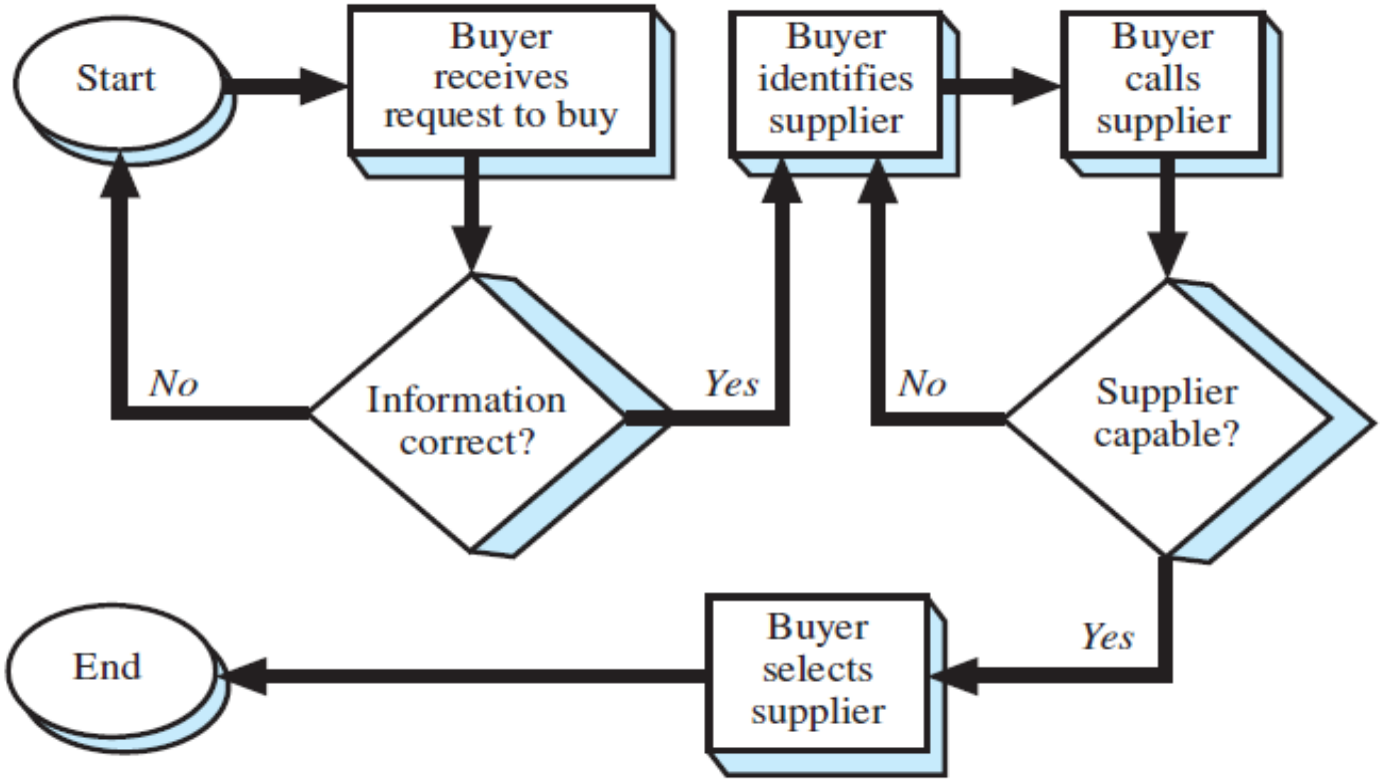


Decision Points

Direction

Flow Chart

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management



Source: Schroeder, Goldstein

Time Function Mapping

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

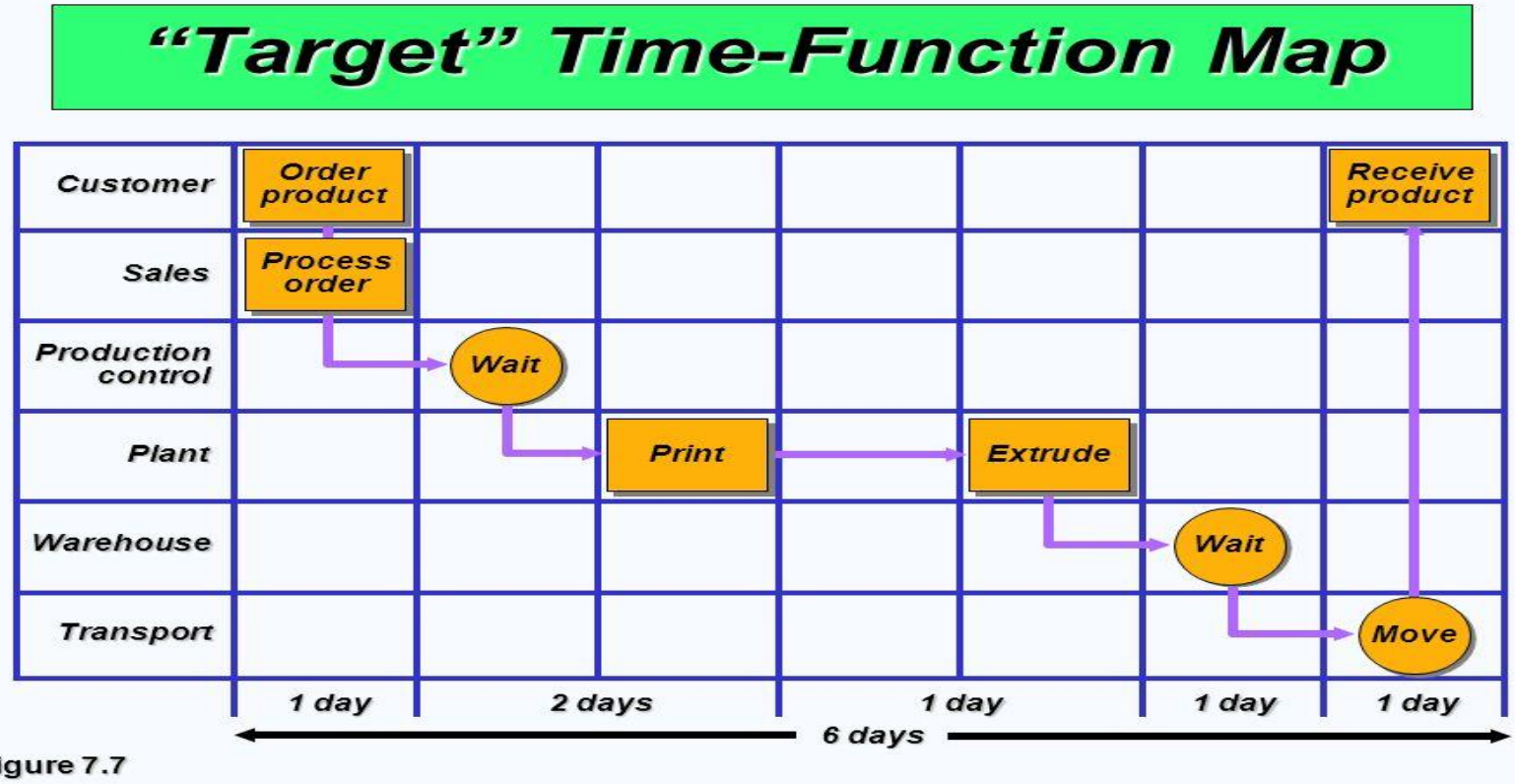


Figure 7.7

© 2008 Prentice Hall, Inc.

A flowchart with time added on the horizontal axis

Process Chart

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

Present Method [X] PROCESS CHART Proposed Method [ ]

SUBJECT CHARTED Hamburger Assembly Process DATE 12 / 1 / 15

DEPARTMENT _____ CHART BY KH SHEET NO. 1 OF 1

DIST. IN FEET	TIME IN MINS.	CHART SYMBOLS	PROCESS DESCRIPTION
	—		Meat Patty in Storage
1.5	.05		Transfer to Broiler
	2.50		Broiler
	.05		Visual Inspection
1.0	.05		Transfer to Rack
	.15		Temporary Storage
.5	.10		Obtain Buns, Lettuce, etc.
	.20		Assemble Order
.5	.05		Place in Finish Rack
3.5	3.15	2 4 1 – 2	TOTALS

Value-added time = Operation time/Total time = (2.50+.20)/3.15 = 85.7%

○ = operation; ⇨ = transport; □ = inspect; D = delay; ▽ = storage.

Analyze the movement of people and material

Source: Heizer, Render, Muckson

Value Stream Mapping

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

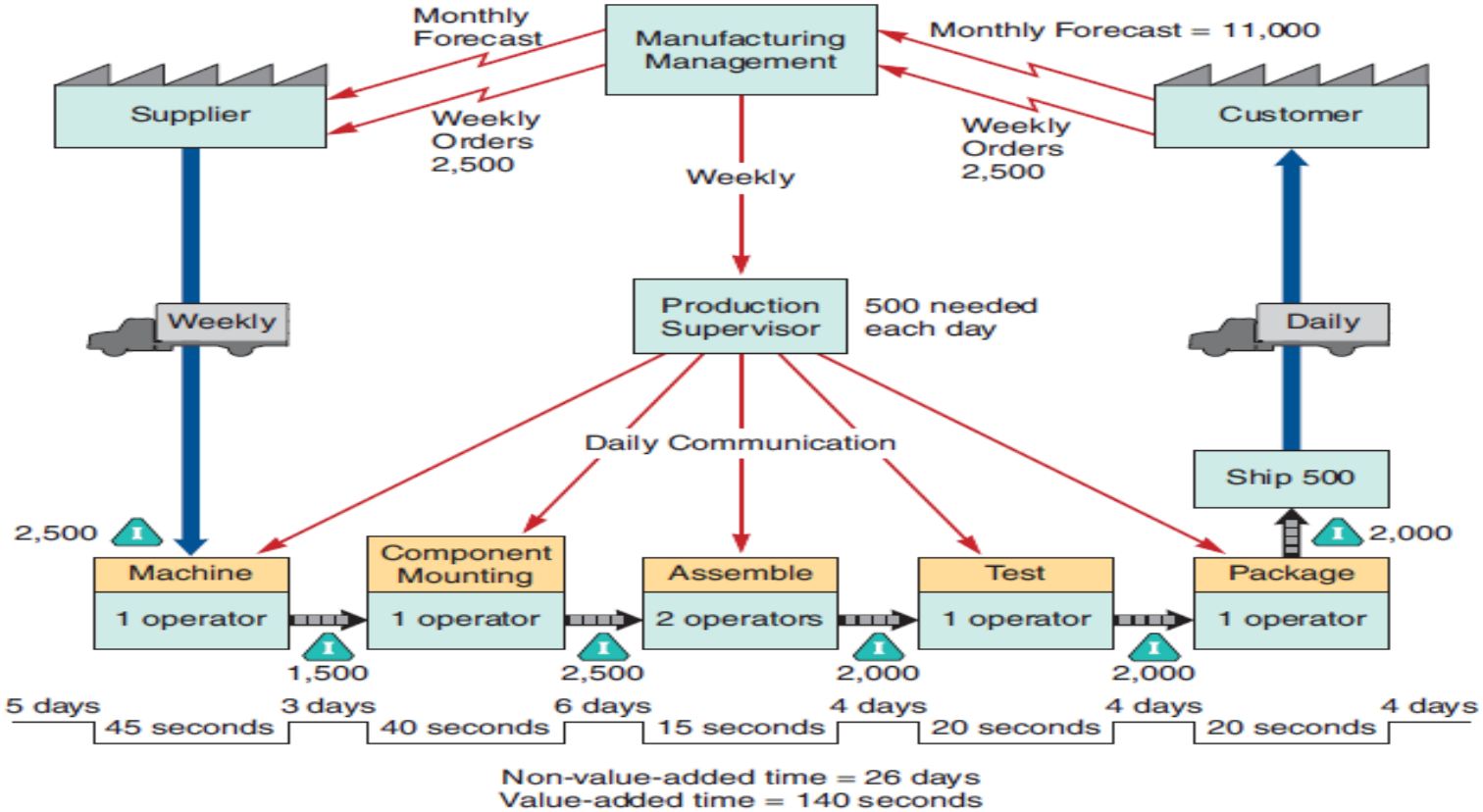
- Value Stream is the stream of activities (both value & non-value added) that are required to provide products / services to the customer.
- Commonly used to identify opportunities for improvement.

There are two types of VSM :

- As-Is Value Stream Map : VSM on “what is observed”. It is also known as Current State Map.
- To-Be Value Stream Map : VSM on “what it should be”. It is also known as Future State Map.

Class Exercise

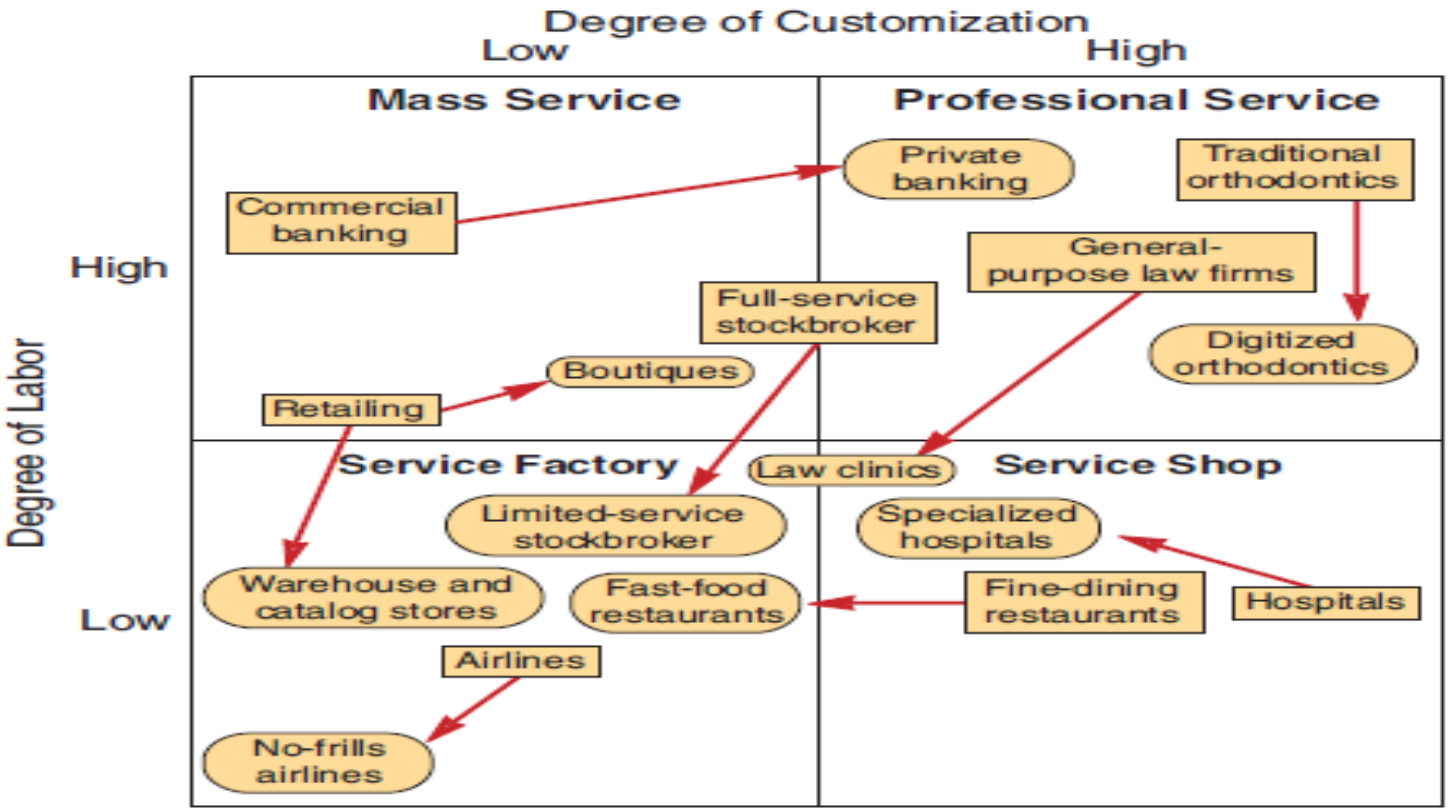
Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management



Source: Heizer, Render, Muckson

Service Process Design

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management



Source: Heizer, Render, Muckson

Service Process Design

- Separation : State Bank Of India, HDFC, ICICI.
- Self Service : More, Reliance Fresh, Spencer.
- Postponement : Asian Paint, Birla Opus.
- Focus : Burger King, KFC.
- Modules : Taparia Tools.
- Automation : Honeywell.
- Scheduling : Vimcal.
- Training: NIIT.

Any Question

Production Technology

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

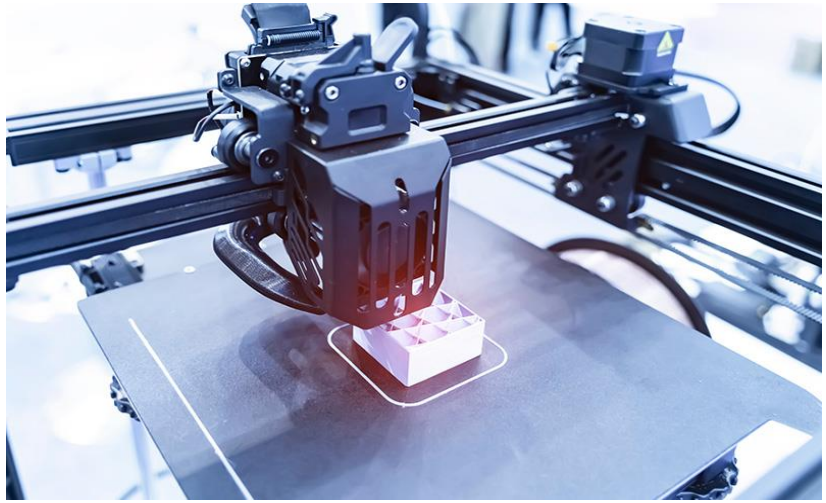
- Technologies used in the production system to improve employee safety, productivity and quality.
- The different types of technologies used are :
- Machine technology, Automatic identification systems (AIS), Process Control, Vision systems, Robots, Automated storage and Retrieval systems (ASRSs), Automated guided vehicles (AGVs), Flexible manufacturing systems (FMSs).

Machine Technology

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

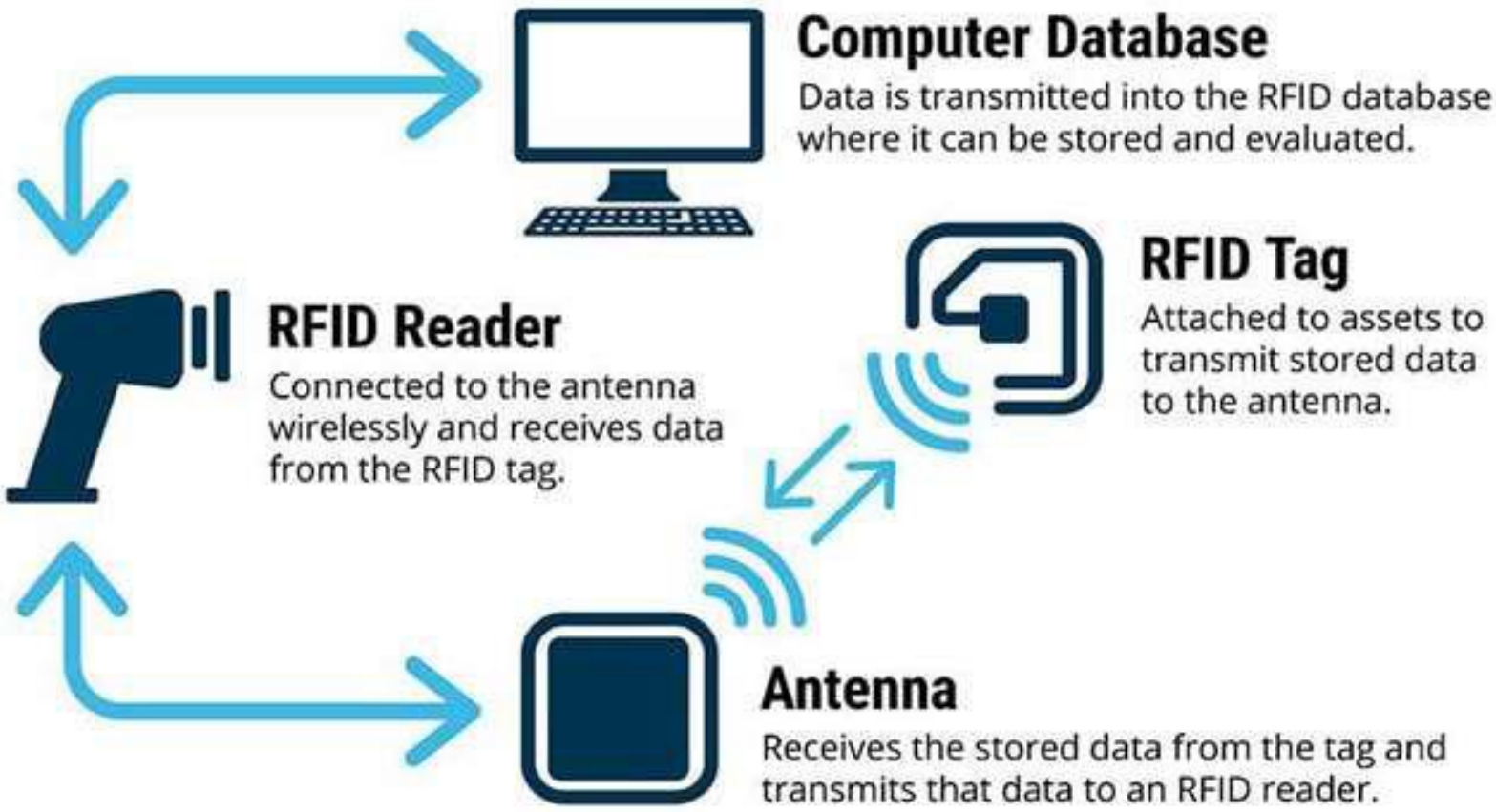
Computer numerical control (CNC)

Additive manufacturing



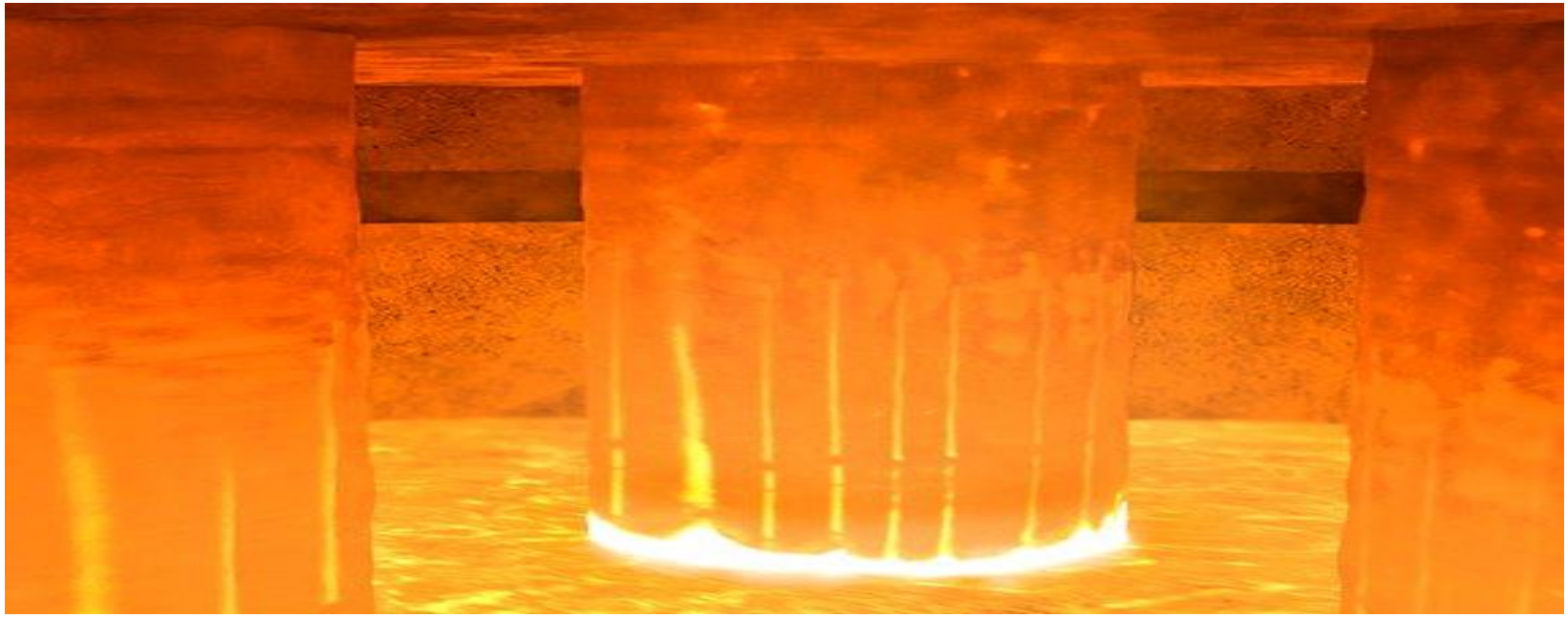
Automatic identification systems (AIS)

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management



Vision System

- Systems that use video cameras and computer technology in inspection roles.



Hot Temperature Camera.

Source: Metso India

Flexible manufacturing system (FMS)

Robot

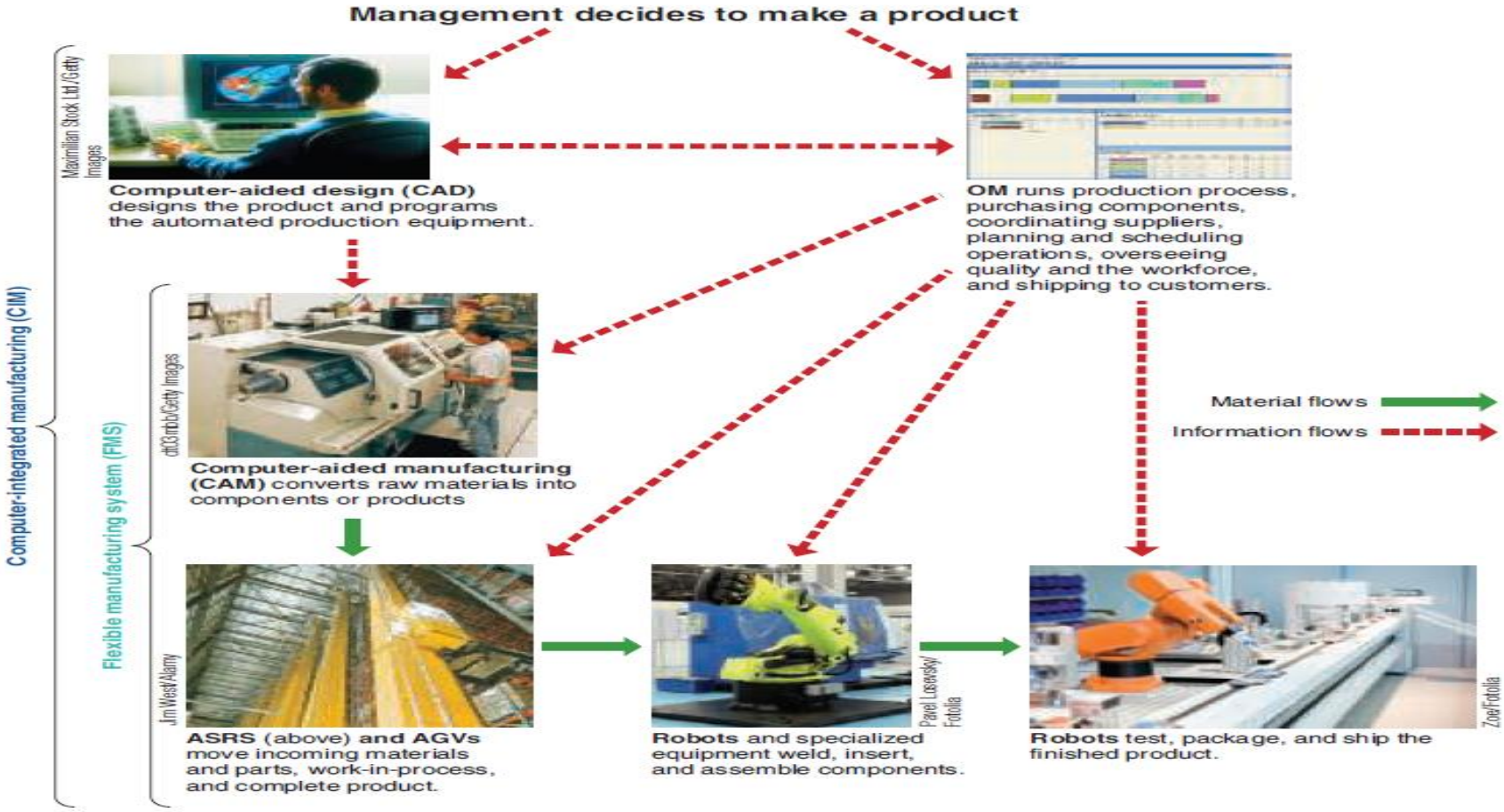
- A flexible machine with the ability to hold, move, or grab items.

ASRS & AGV

- **Automated storage and retrieval systems (ASRS)** — Computer-controlled warehouses that provide for the automatic placement of parts into and from designated places within a warehouse.
- **Automated guided vehicle (AGV)** —Electronically guided and controlled cart used to move materials.

Computer-integrated manufacturing (CIM)

- CIM includes computer-aided design (CAD), computer-aided manufacturing (CAM), flexible manufacturing systems (FMSs), automated storage and retrieval systems (ASRSs), automated guided vehicles (AGVs), and robots to provide an integrated and flexible manufacturing process.



Source: Heizer, Render, Muckson

Industry 4.0

- https://www.youtube.com/watch?v=h4Bwj8hqTJE

Technology in Service Sector

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

TABLE 7.4 Examples of Technology's Impact on Services

SERVICE INDUSTRY	EXAMPLE
Financial Services	Debit cards, electronic funds transfer, automatic teller machines, Internet stock trading, online banking via cell phone
Education	Online newspapers and journals, interactive assignments via WebCT, Blackboard, and smartphones
Utilities and government	Automated one-person garbage trucks, optical mail scanners, flood-warning systems, meters that allow homeowners to control energy usage and costs
Restaurants and foods	Wireless orders from waiters to the kitchen, robot butchering, transponders on cars that track sales at drive-throughs
Communications	Interactive TV, e-books via Kindle
Hotels	Electronic check-in/check-out, electronic key/lock systems, mobile Web bookings
Wholesale/retail trade	Point-of-sale (POS) terminals, e-commerce, electronic communication between store and supplier, bar-coded data, RFID
Transportation	Automatic toll booths, satellite-directed navigation systems, Wi-Fi in automobiles
Health care	Online patient-monitoring systems, online medical information systems, robotic surgery
Airlines	Ticketless travel, scheduling, Internet purchases, boarding passes downloaded as two-dimensional bar codes on smartphones

Source: Heizer, Render, Muckson

Any Question

Take Home Exercise

- Read the Case let " Repetitive Manufacturing Works at Harley Davidson" of your Text Book.
- Service Blue Printing.

Further Reading / References

1. **Operation Management by Jay Heizer, Barry Render & Chuck Munson—Chapter 7**
2. Operation Management by William J Stevenson—Chapter 6
3. Operations Management by Russell and Taylor—Chapter 6

Prepared by Prof. Abhishek Maiti for the use of the students of Operation Management

THANK YOU